Summer 1983

Problem 1 (Su83). The number 2198214591730833048701336 9 is the thirteenth power of a positive integer. Which positive integer?

Problem 2 (Su83). Let $f : \mathbb{C} \to \mathbb{C}$ be an analytic function such that

$$\left(1 + |z|^k\right)^{-1} \frac{\mathrm{d}^m f}{\mathrm{d}z^m}$$

is bounded for some k and m. Prove that $\mathrm{d}^n f/\mathrm{d}z^n$ is identically zero for sufficiently large n. How large must n be, in terms of k and m?

Problem 3 (Su83). Let A be an $n \times n$ complex matrix, and let χ and μ be the characteristic and minimal polynomials of A. Suppose that

$$\chi(x) = \mu(x)(x - i)$$

$$\mu(x)^2 = \chi(x)(x^2 + 1)$$

Determine the Jordan Canonical Form of A.

Problem 4 (Su83). Outline a proof, starting from basic properties of the real numbers, of the following theorem: Let $f : [a, b] \to \mathbb{R}$ be a continuous function such that $f'(x) = 0$ for all $x \in (a, b)$. Then $f(b) = f(a)$.

Problem 5 (Su83). Let $b_1, b_2, \ldots$ be positive real numbers with

$$\lim_{n\to\infty} b_n = \infty \qquad \text{and} \qquad \lim_{n\to\infty} (b_n/b_{n+1}) = 1$$

Assume also that $b_1 < b_2 < b_3 < \cdots$. Show that the set of quotients $\left(\frac{b_m}{b_n}\right)_{1\leqslant n<m}$ is dense in $(1, \infty)$.

Problem 6 (Su83). Let V be a real vector space of dimension n with a positive definite inner product. We say that two ordered bases (a_n) and (b_n) have the same *orientation* if the matrix of the change of basis from (a_n) to (b_n) has a positive determinant. Suppose now that (a_n) and (b_n) are orthonormal bases with the same orientation. Show that $(a_n + 2b_n)$ is again a basis of V with the same orientation as (a_n).

Problem 7 (Su83, Sp84, Fa89). Compute $\displaystyle\int_0^\infty \frac{\log x}{x^2 + a^2}\,\mathrm{d}x$ where $a > 0$ is a constant.

Problem 8 (Su83). Let G_1, G_2, and G_3 be finite groups, each of which is generated by its commutators, that is, elements of the form $xyx^{-1}y^{-1}$. Let A be a subgroup of $G_1 \times G_2 \times G_3$, which maps surjectively, by the natural projection map, to the partial products $G_1 \times G_2$, $G_1 \times G_3$ and $G_2 \times G_3$. Show that A is equal to $G_1 \times G_2 \times G_3$.

Problem 9 (Su83). Suppose Ω is a bounded domain in $\mathbb{C}$ with a boundary consisting of a smooth Jordan curve γ. Let f be holomorphic on a neighborhood of the closure of Ω, and suppose that $f(z) \neq 0$ for $z \in \gamma$. Let $z_1, \ldots, z_k$ be the zeros of f in Ω, and let n_j be the order of the zero of f at z_j (for $j = 1, \ldots, k$).

1. Use Cauchy's integral formula to show that $\dfrac{1}{2\pi i} \displaystyle\int_\gamma \frac{f'(z)}{f(z)}\, dz = \sum_{j=1}^{k} n_j$
2. Suppose that f has only one zero z_1 in Ω with multiplicity $n_1 = 1$. Find a boundary integral involving f whose value is the point z_1.

Problem 10 (Su83). Let f be a twice differentiable real valued function on $[0, 2\pi]$, with $\int_0^{2\pi} f(x)\, dx = 0 = f(2\pi) - f(0)$. Show that

$$\int_0^{2\pi} (f(x))^2\, dx \leqslant \int_0^{2\pi} \left(f'(x)\right)^2 dx$$

Problem 11 (Su83). Find the eigenvalues, eigenvectors, and the Jordan Canonical Form of

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

considered as a matrix with entries in $\mathbb{F}_3$.

Problem 12 (Su83, Fa18). Prove that a continuous function from $\mathbb{R}$ to $\mathbb{R}$ which maps open sets to open sets must be monotonic.

Problem 13 (Su83). Let A be an $n \times n$ complex matrix, all of whose eigenvalues are equal to 1. Suppose that the set $\{A^n \mid n = 1, 2, \ldots\}$ is bounded. Show that A is the identity matrix.

Problem 14 (Su83). Let G be a transitive subgroup of S_n. Suppose that G is a simple group and that $\sim$ is a *G-invariant* equivalence relation on $\{1, \ldots, n\}$, that is, $i \sim j$ implies that $\sigma(i) \sim \sigma(j)$ for all $\sigma \in G$. What can one conclude about such G-invariant relations?

Problem 15 (Su83, Fa96). Let f be analytic on and inside the unit circle $C = \{z \mid |z| = 1\}$. Let L be the length of the image of C under f. Show that $L \geqslant 2\pi|f'(0)|$.

Problem 16 (Su83). Let Ω be an open subset of $\mathbb{R}^2$, and let $f : \Omega \to \mathbb{R}^2$ be a smooth map. Assume that f preserves orientation and maps any pair of orthogonal curves to a pair of orthogonal curves. Show that f is holomorphic with the identification $\mathbb{R}^2 \cong \mathbb{C}$.

Problem 17 (Su83, Sp10). Let A be an $n \times n$ Hermitian matrix satisfying the condition $A^5 + A^3 + A = 3I$. Show that $A = I$.

Problem 18 (Su83). Find all real valued C^1 solutions $y(x)$ of the differential equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + y = x \qquad -1 < x < 1$$

Problem 19 (Su83, Sp14). Compute the area of the image of the unit disc $\mathbb{D} = \{z \mid |z|<1\}$ under the map $f(z) = z + z^2/2$.

Problem 20 (Su83). Let $f : \mathbb{R} \to \mathbb{R}$ be continuously differentiable, periodic with period 1, and nonnegative. Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f(x)}{1 + cf(x)}\right) \to 0 \quad \text{as } c \to \infty$$

uniformly in x.